

Appendix A - Description of test areas at Andøya



- RED** = Official Test Area 1
- Green** = Official Test Area 2
- Blue** = Official Test Area 3

Survey points

Notice: Geodetic reference frame for plane coordinates is EUREF89.

Differences between EUREF89 and WGS84, see Appendix H.

Seven significant decimal digits for latitude and longitude will ensure cm-precision.

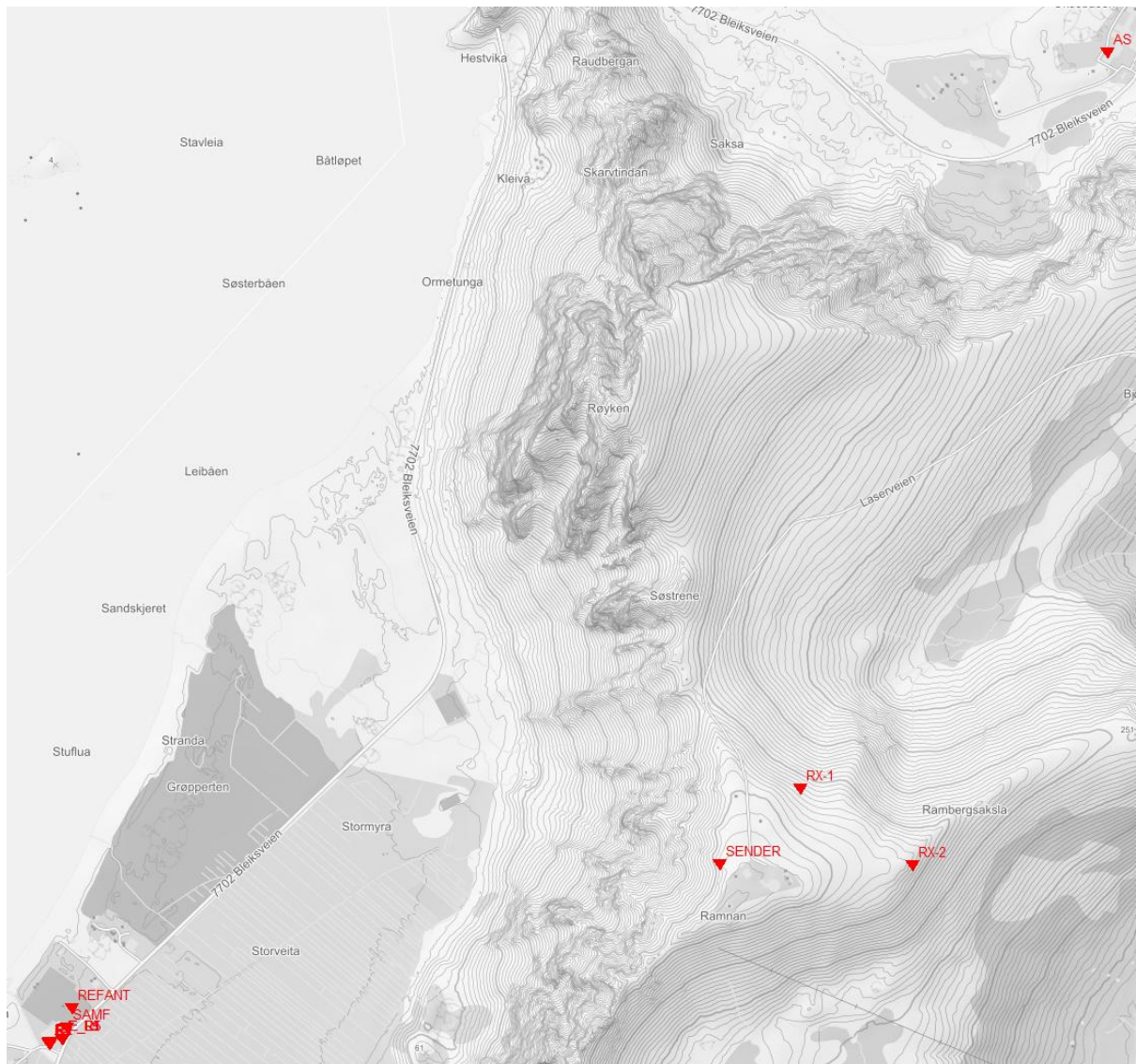
Table 1: Coordinates for Jammertest 2026 (Physical height = heights in The Norwegian NN2000 height system)

Point-ID	UTM33-North	UTM33-East	Physical height	Latitude	Longitude	Ellipsoidal height	Mark
ORIG	7679455,50	536952,70	31,33	69,2225166	15,9334241	67,22	Origo Site2
SENDER	7685781,10	539724,98	361,49	69,2788429	16,0060623	397,28	FFI Jammer
RX-1	7685948,78	539904,09	315,76	69,2803199	16,0106675	351,54	Antenna FFI
RX-2	7685779,70	540153,98	322,14	69,2787668	16,0169243	357,90	Antenna FFI
REFANT	7685462,32	538282,87	3,12	69,2761930	15,9694187	38,97	Antenna
SAMF	7685417,69	538272,09	3,46	69,2757944	15,9691279	39,31	Lift antenna
R1	7685386,74	538234,19	3,86	69,2755222	15,9681560	39,71	Corner antenna rig
R2	7685384,16	538236,05	3,75	69,2754988	15,9682020	39,60	Corner antenna rig
R3	7685382,32	538233,40	3,74	69,2754826	15,9681343	39,59	Corner antenna rig
R4	7685384,89	538231,65	3,81	69,2755059	15,9680910	39,66	Corner antenna rig
MEAC1	7680854,71	537417,51	62,71	69,2349991	15,9457087	98,58	Antenna FOI
MEAC2	7680851,60	537421,29	62,70	69,2349707	15,9458028	98,57	Antenna FOI
A50	7679505,41	536955,45	29,97	69,2229637	15,9335127	65,85	Tree stick top
A100	7679555,31	536958,19	29,85	69,2234108	15,9336011	65,74	Tree stick top
A150	7679605,22	536960,94	29,89	69,2238580	15,9336898	65,77	Tree stick top
B50	7679428,17	536994,55	30,19	69,2222658	15,9344705	66,07	Tree stick top
B100	7679400,84	537036,39	31,34	69,2220150	15,9355167	67,22	Tree stick top
B150	7679373,51	537078,24	33,54	69,2217642	15,9365630	69,42	Tree stick top
C50	7679432,93	536908,11	29,17	69,2223203	15,9322893	65,05	Tree stick top
C100	7679410,35	536863,52	28,49	69,2221239	15,9311544	64,38	Tree stick top
C150	7679387,78	536818,92	28,16	69,2219276	15,9300194	64,05	Tree stick top
D50	7679478,07	536997,29	30,90	69,2227129	15,9345589	66,79	Tree stick top
D100	7679500,65	537041,89	31,95	69,2229092	15,9356941	67,83	Tree stick top
D150	7679523,22	537086,48	34,86	69,2231055	15,9368289	70,74	Tree stick top
E50	7679405,60	536949,95	30,14	69,2220694	15,9333354	66,02	Tree stick top
E100	7679355,69	536947,21	29,25	69,2216223	15,9332470	65,13	Tree stick top
E150	7679305,78	536944,46	29,96	69,2211752	15,9331584	65,85	Tree stick top
F50	7679482,83	536910,85	29,02	69,2227673	15,9323776	64,91	Tree stick top
F100	7679510,16	536869,01	28,22	69,2230181	15,9313314	64,11	Tree stick top
F150	7679537,49	536827,16	27,81	69,2232689	15,9302849	63,70	Tree stick top
N500	7679955,50	536952,70	30,00	69,2270000	15,9336165	65,89	«Help point»
301	7670668,62	530198,75	8,41	69,1445605	15,7600882	44,52	Painted mark
302	7670680,37	530212,30	8,49	69,1446644	15,7604330	44,61	Painted mark
303	7677529,28	534164,53	8,17	69,2056106	15,8623208	44,16	Painted mark
304	7677547,58	534161,78	8,09	69,2057750	15,8622578	44,08	Painted mark
305	7678211,00	534007,45	6,02	69,2117434	15,8585975	42,02	Painted mark
306	7678203,81	534010,27	6,11	69,2116786	15,8586663	42,10	Painted mark
JAM-S3A	7678283,66	533997,48	5,33	69,2123962	15,8583718	41,33	Painted mark
JAM-S3B	7678283,26	533994,81	5,38	69,2123930	15,8583042	41,37	Painted mark
307	7679014,60	534010,65	6,43	69,2189489	15,8589629	42,43	Painted mark
308	7679027,13	534017,55	6,21	69,2190604	15,8591415	42,21	Painted mark
309	7677843,92	534179,92	6,87	69,2084301	15,8628211	42,86	Painted mark
310	7677829,74	534174,75	6,96	69,2083036	15,8626854	42,95	Painted mark
AS	7687584,32	540587,21	7,91	69,2948831	16,0286613	43,66	Antenna

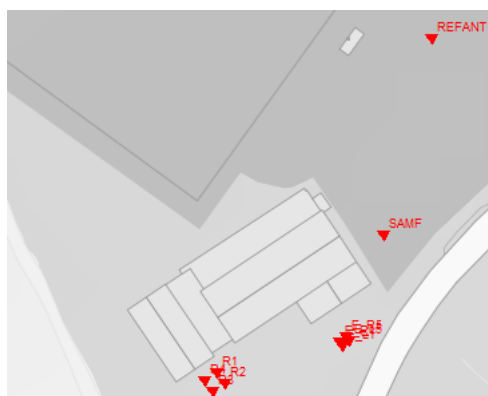
E_R1	7685393,59	538262,01	6,51	69,2755797	15,9688632	42,36	Ericsson
E_R5	7685394,71	538263,62	6,50	69,2755894	15,9689043	42,35	Ericsson
E_L5	7685393,89	538264,16	6,50	69,2755820	15,9689175	42,35	Ericsson
E_L1	7685392,77	538262,63	6,50	69,2755722	15,9688784	42,35	Ericsson

Description of Test Area 1

Overview maps of survey points



Scale 1:15000



Scale 1:2000 More detailed view of surveyed points outside Bleik samfunnshus.

Description of Test Area 2

Test Area 2 is the parking area at the end of a dirt road. (Position N 69.2225°, Ø 15.9335°).

Most of the testing will be conducted at the parking or the surrounding area.

The setup is based around known positions, distances and controlled RF power levels for the tests.

Barracks and parking plan 2026

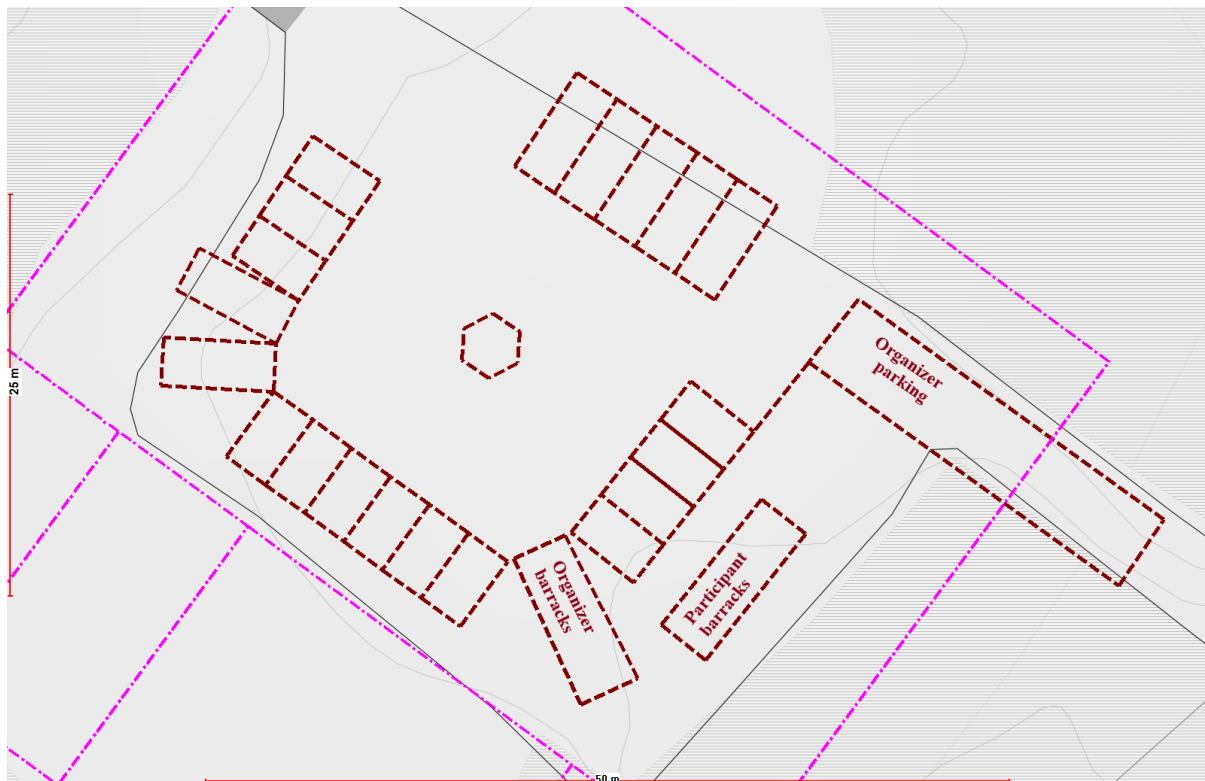


Figure 1: Barracks and parking plan 2026

Points Bauktuva parking area



Scale 1:2000 Points by the barrack

Drone areas 2026

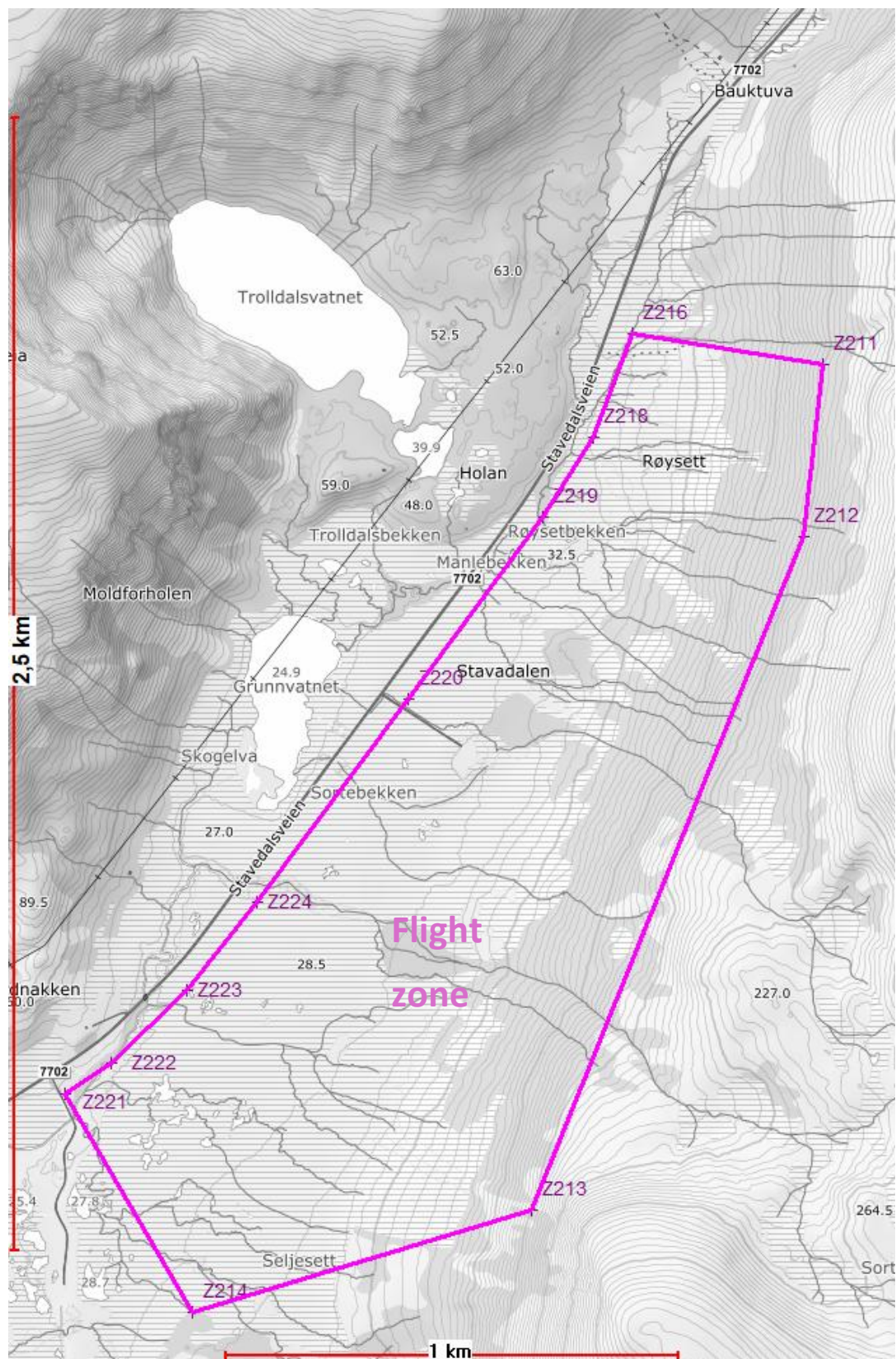


Figure 2: Drone area boundary

Jammer and drone markers 2026

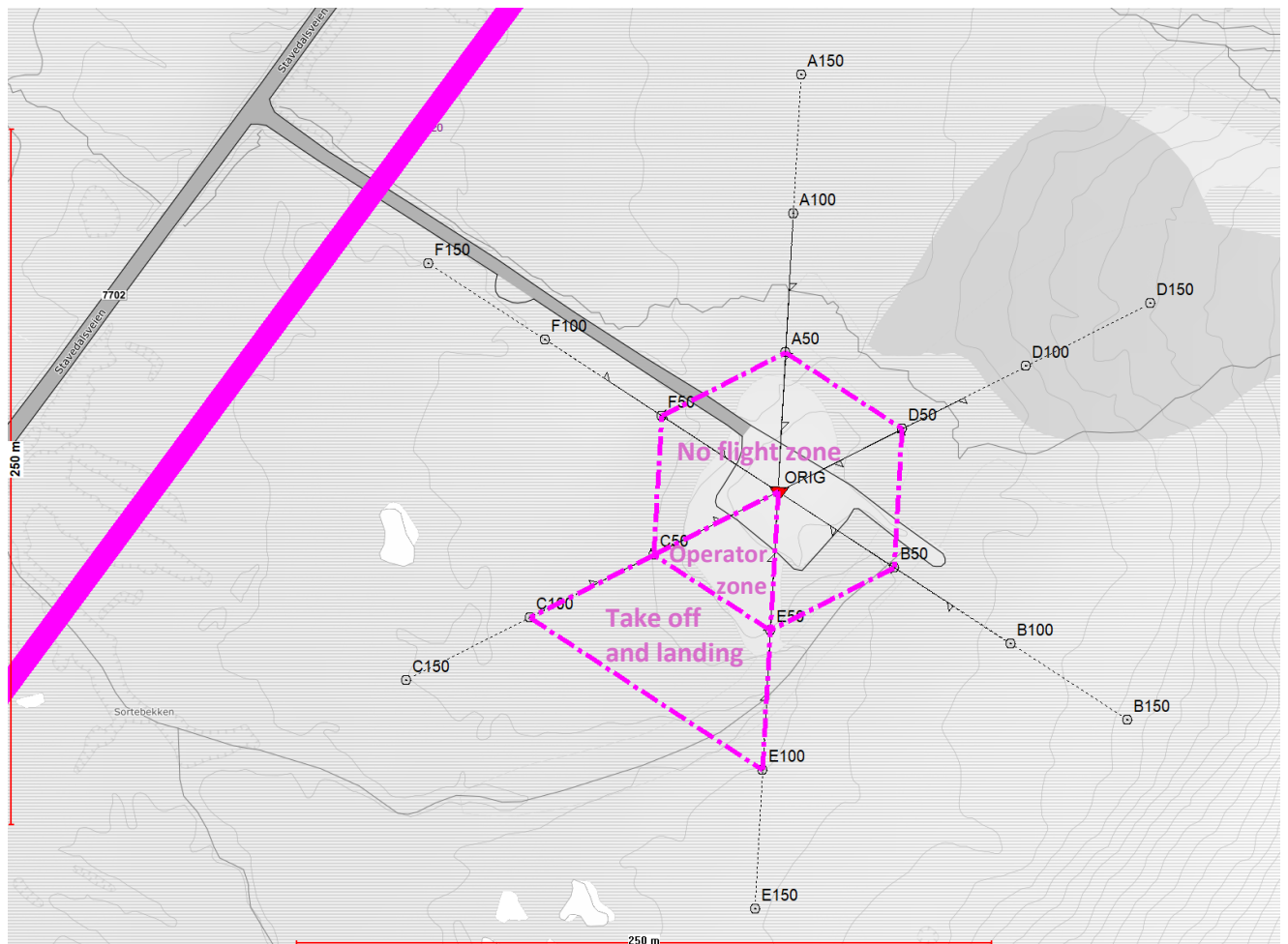


Figure 3 Test Area 2, directions for where we place the jammers and several new directions (2026) that will work as visual markers from the air as “drone waypoints”.

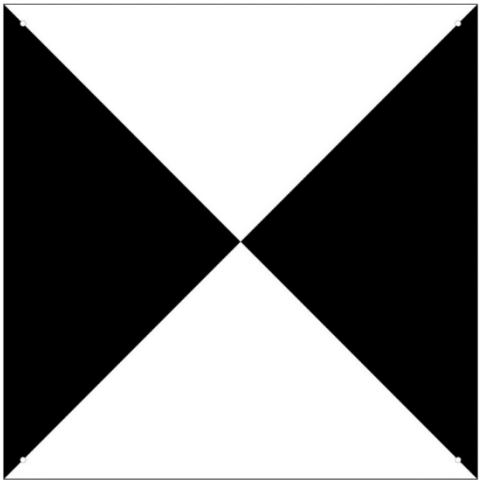
The test setup at location 2

A, B and C axis are positions used for placement of jammers, or spoofing equipment, as shown in figure 3. A, B and C axis is separated 120 degrees apart. Points will be marked with “wood sticks” (trelekter).

D, E and F axis are continuations of C, A and B axis. They will be marked with drone waypoints visual from the air.

D, E and F- axis will have black and white drone markers, which should be visible at 100 meters above ground level.

Example: drone marker/ visual waypoint. (Dimensions 50 cm x 50 cm)



Fiducial marks added to drone markers for Test Area 2.

Axis marked with color and distance marked with squares.

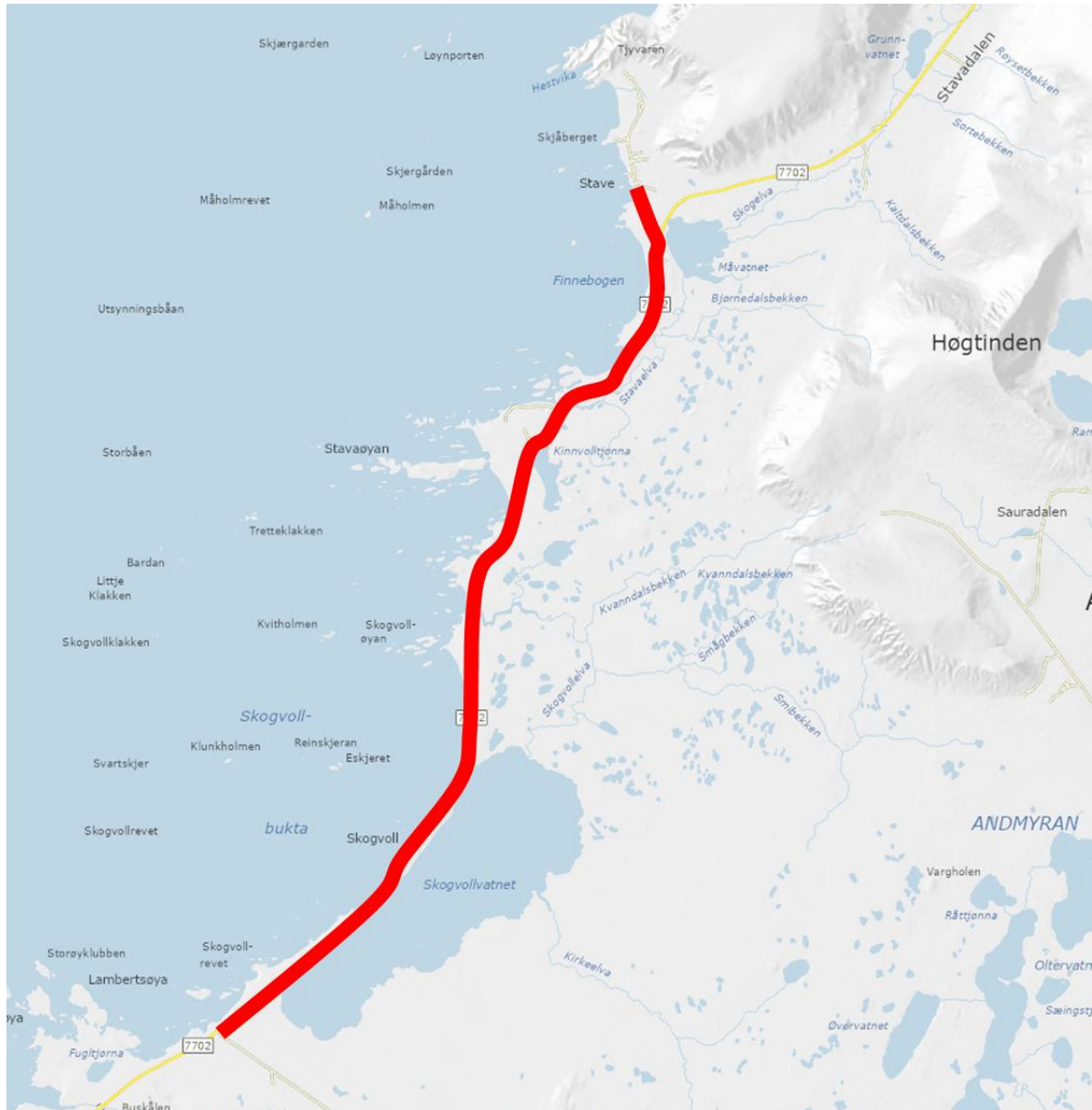
Red	=	D - direction
Blue	=	E - direction
Green	=	F - direction
1 square	=	50 meters
2 squares	=	100 meters
3 squares	=	150 meters

Example: D – direction @ 150 meters.



Description of motorcade route(s) on Andøya, Test Area 3

The start is at Stave community house (69.212187 North, 15.858559 East), the small jammers can be used at the intersection between county road 7702 and communal road "Okleveien" (69.14409 North, 15.75847 East). The figure below shows the stretch that can be used for the motorcade (Red line).



The road is quite narrow 5.1 meters with a speed limit of 80 km/h. The traffic volume is low with about 1000 vehicles per day. For some tests where reduced speed is needed there will be a NPRA vehicle in front and at the back of motorcade. Communication to the vehicles will be via FM radio.

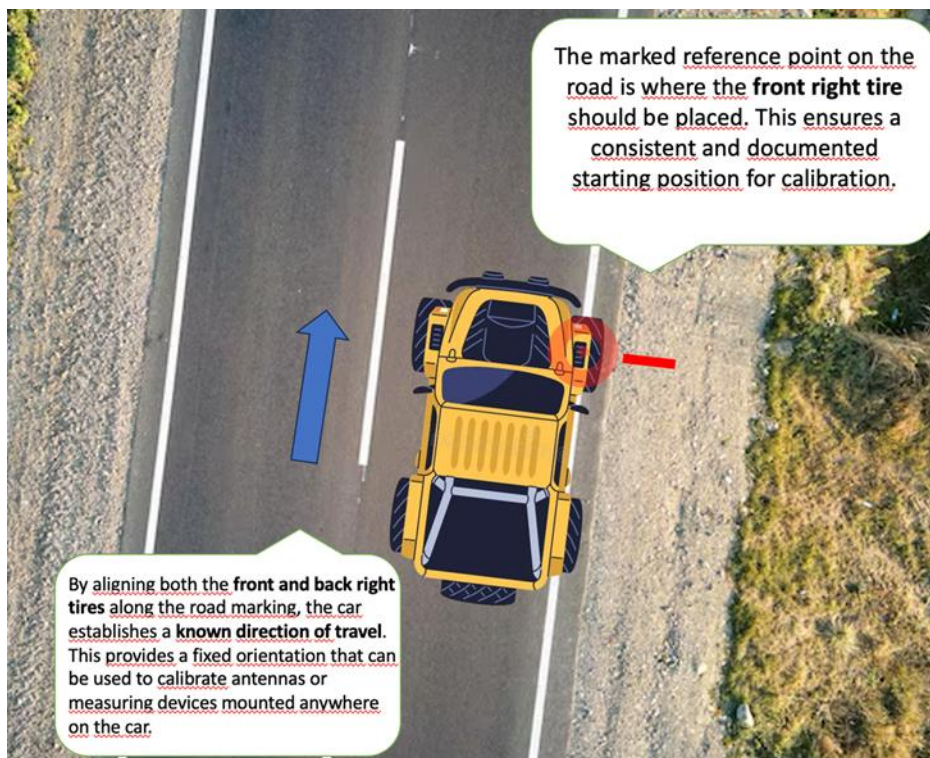
Calibration Control Marks

In a straight, open road section, three pairs of control marks (301-310) will be painted in center of the dotted white line at the right-hand side. The marks will be positioned so that vehicles can align with them as follows:

- The **front right wheel** is placed directly on the painted mark.
- The **rear right wheel** is aligned with the center of the white line.

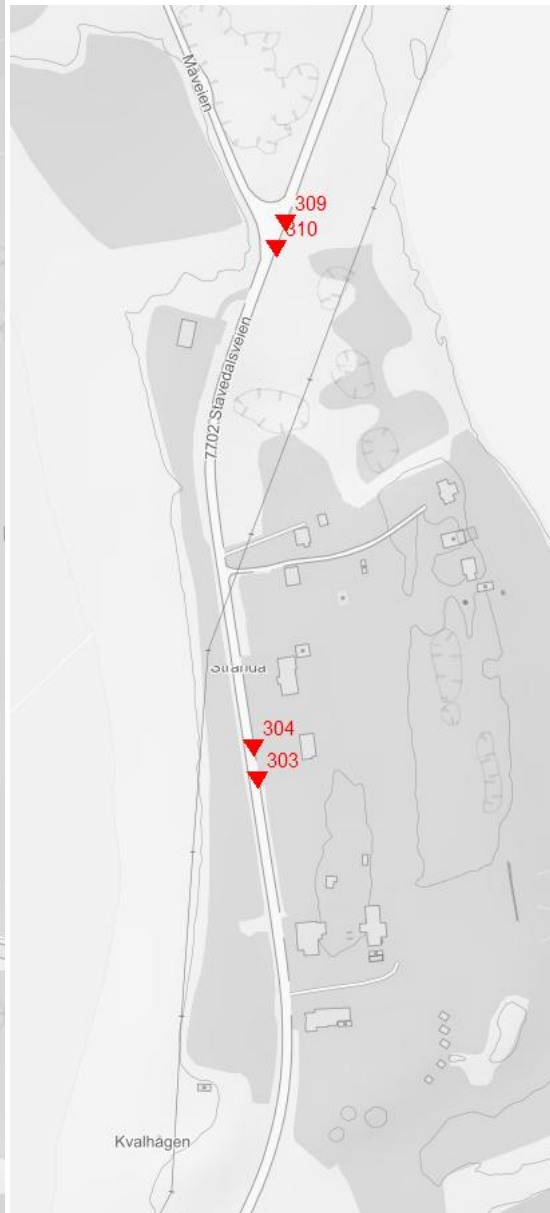
This arrangement provides a precise and repeatable reference for the forward right wheel position relative to the vehicle's facing direction.

The coordinates of the control marks will be averaged from several time independent RTK-measurements and distributed in *Appendix A Table 1*. These marks are intended for calibrating on-board GNSS and inertial navigation equipment against a known reference point.

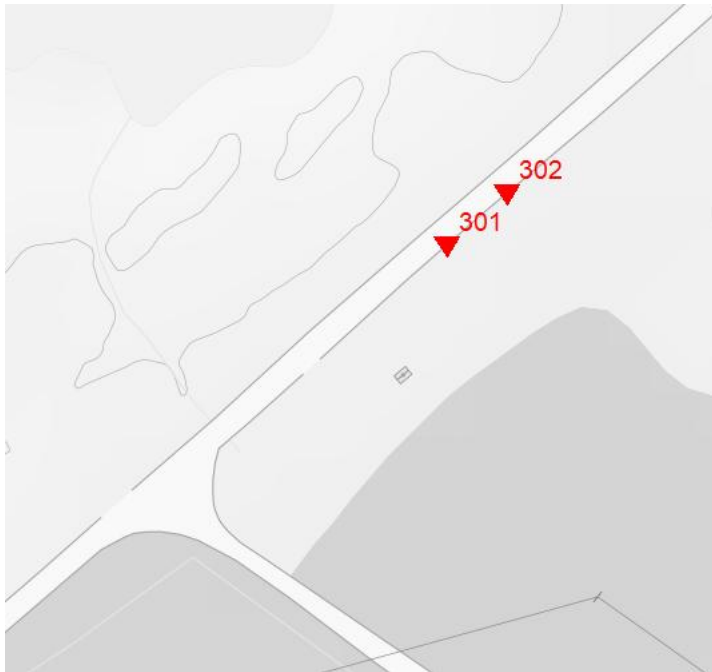




Scale 1:50000



Scale 1:5000



Scale 1:2000 Example, one pair of calibration points